

Kakhovka dam breach leaves thousands without water, threatens summer crops

Key Messages

- On June 6, the Kakhovka dam on the Dnipro River was damaged by the conflict, flooding an estimated **410-420 square kilometers of land** in the Kherson region, according to NASA Harvest analysis. The flooding affected **over 20 villages and cities**, reportedly temporarily displacing **an estimated 4,000 people** and leaving thousands without electricity and water. **Humanitarian organizations** responded immediately with **food**, water, cash assistance, and other supplies. However, as of late June, **thousands** of Ukrainians in Kherson still do not have access to clean drinking water and are relying on water tanks. Additionally, an estimated 210,000 people in Dnipropetrovsk *oblast* do not have sufficient access to clean water due to rapidly declining water levels in the Kakhovka Reservoir.
- The Kakhovka dam disaster comes on top of mounting concern for the **contamination of Ukraine's public water supplies** with heavy metals and other toxins (likely including radioactive sediments) in the course of the conflict, including due to the dumping of military equipment and materials in rivers, damage to oil and industrial facilities, and the flooding of mines due to power outages. The UN estimates that **16 million people** in Ukraine required water, sanitation, and hygiene assistance from April to December 2022.
- The direct damage to cropland following the Kakhovka dam was comparatively minor. According to NASA Harvest estimates, approximately **3.5-5 square kilometers of cropland** were damaged by the flooding. While **global wheat prices rose** as the market overreacted to the disaster, they quickly **stabilized** in subsequent days. However, the depletion of the Kakhovka Reservoir and consequent disconnection of four canals from the reservoir as of late June is raising **concerns for the future of irrigated agriculture** – important during the summer growing period – in southern Ukraine. The four affected canals provided water to cropland that produced about **2 million tons of grains and oilseeds** in 2021 (before the conflict). That year, Ukraine harvested a record **86 million tons of grain** alone across the country, suggesting that the share of national production from the region served by the four canals was small but non-negligible. In recent years, the share of national crop production in the southern region has **declined** due to impacts of the conflict. As such, the identified future challenges in irrigation capacity are of greatest concern for local agricultural livelihoods and income-earning rather than for national- or global-level impacts of reduced agricultural production levels.
- To date, a key Russian **demand** during negotiations around the extension of the Black Sea Safe Grain Corridor deal has been the resumption of exports of Russian ammonia (an important ingredient in nitrogen fertilizer) via the Togliatti-Odessa pipeline. However, on June 5, the pipeline suffered **damage** that will reportedly take **one to three months** to repair. As such, Russia has lost **prospects of immediate gain** from agreeing to extend the Safe Grain Corridor deal. Meanwhile, media reports indicate that a key Russian producer of ammonia has identified an alternative export route. However, in May, the CEO of the company reportedly stated that the first stage of construction needed for exports via this new route could be **completed by the end of the year**, suggesting that the full completion of the project will take substantial time, although information is limited.
- In late June, Ukrainian officials have **expressed concern** that Russia will most likely refuse to extend the Black Sea Safe Grain Corridor deal beyond the upcoming expiration date of July 18. Meanwhile, Russia continues to obstruct the deal even before its expiration. According to UN data reported by Reuters, no ships had entered Pivdennyi port for more than three weeks as of June 7. In anticipation of the potential collapse of the deal, Ukrainian officials have established a **547 million USD insurance fund** to incentivize shipping companies to continue operations. However, some speculate that this will not be sufficient protection given the associated

risks and uncertainty. In addition to the insurance fund, Ukrainian officials have indicated that they hope to prioritize the **expansion of export capacity of Danube River ports** by deepening the Bystre Canal. In May, Ukraine exported a record **3 million tons of food** via three Danube River ports.

- **Global wheat prices spiked** in June, in part attributed to the **impacts of the Wagner revolt** in Russia amid concerns for reduced production in key global suppliers. However, concerns eased at the end of the month after the situation calmed in Russia and rain was received in the midwestern United States. With harvesting imminent in the northern hemisphere, global wheat prices will continue to be influenced by **numerous factors** related to supply, demand, and global financial markets. Should the Safe Grain Corridor Deal be allowed to expire, global grain prices will likely rise temporarily as the market reacts, but overall price trends – even the short term – will likely continue to be primarily determined by factors unrelated to Ukraine.

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Key Message Update

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